On my honor, I have neither given nor received any unauthorized aid for inappropriate assistance for all sections of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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1)

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ \frac{3\alpha}{5} & 1 & \frac{-9\gamma}{5} & \frac{1}{5} \\ \frac{2}{3} & \frac{\beta}{3} & 1 & \frac{\alpha^2}{3} \\ \frac{1}{2} & 7 & -3 & 1 \end{bmatrix}$$

$$\frac{\beta}{3} - \frac{2}{3} = -1$$

$$7 - \frac{1}{2} = k$$

$$\beta - 2 = -3$$

$$\boxed{\beta = -1}$$

$$\boxed{k = \frac{13}{2}}$$

$$\frac{2-\beta}{3}$$

$$\frac{1}{2} - 7$$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & ? & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \begin{matrix} -3\alpha\, R_1 + R_2 \to R_2 \\ -2R_1 + R_3 \to R_3 \\ \longrightarrow \\ -R_1 + R_4 \to R_4 \end{matrix} \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5-3\alpha^2 & 9\alpha-9\gamma & 1-3\alpha^2 \\ 0 & \beta-2\alpha & 2\gamma+3 & \alpha^2-2\alpha \\ 0 & 14-\alpha & \gamma-6 & 2-\alpha \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 1 & \frac{9\alpha-9\gamma}{5-3\alpha^2} & \frac{1-3\alpha^2}{5-3\alpha^2} \\ 0 & \beta-2\alpha & 2\gamma+3 & \alpha^2-2\alpha \\ 0 & 14-\alpha & \gamma-6 & 2-\alpha \end{bmatrix}$$

2)

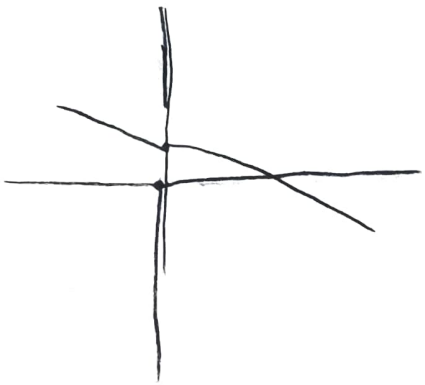
a) $x = 5y$ $\qquad y = \frac{x}{5}$

$x - 5y = 0$

$$\begin{bmatrix} 1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

one vector in two dimensional space the minimum k is 2

b)



this intersection cant have a volume because it is two dimensional only x and y we need at least three values.

c)

$x + 4y - z = 6$

$$\begin{bmatrix} 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 6$$

x is basic

y and z is free

$y = a$ $\qquad x + 4a - b = 6$

$z = b$ $\qquad x = 6 - 4a + b$ $\qquad S.S = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix} + a \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} b + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

They dont form a vector space they are intersected

1 dimensional in 3 dimension.

(5)

3) $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $n = 2$ in R^2

2×2 $H = I - 2uu^T$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot [1\ 0]$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} [1\ 0]$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = H$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = H^2$$

a)

$v = \begin{bmatrix} 1 \\ \vdots \\ 0 \end{bmatrix}$ in n dimension

$$H = I - 2 \begin{bmatrix} 1 \\ \vdots \\ 0 \end{bmatrix} \cdot [1 \cdots 0]$$

$$2 \cdot \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & & \\ 0 & \cdots & 0 \end{bmatrix} = \begin{bmatrix} 2 & \cdots & 0 \\ \vdots & & \\ 0 & \cdots & 0 \end{bmatrix} = \begin{bmatrix} -1 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 1 \end{bmatrix} = H$$

$I -$

$$H^2 = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 1 \end{bmatrix} = I$$

b) H is an Identity matrix and its orthonormal

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

$\sqrt{1-0} = 1$ the length is 1 and its orthonormal.

c) $H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$H - \lambda I$

$$\begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0$$

$(1-\lambda)^2 = 0$

$\lambda = 1$

Only one lambda

d) H is an Identy matrix and it is already diagonalize.

4) gradobi tork 30 goni
i wiesi ile obocek w Szwecji

$$V_{k+1} = V_k - 3V_k + W_k \quad \Rightarrow V_{k+1} = -2V_k + W_k$$

$$W_{k+1} = W_k - 3W_k + V_k \quad \Rightarrow W_{k+1} = V_k - 2W_k$$

$$\begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} V_k \\ W_k \end{bmatrix} \qquad A - \lambda I$$

$$\begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} = 0$$

$$(-2-\lambda)^2 - 1 = 0$$

$$-2-\lambda = 1 \quad \lambda = -1$$

$$-2-\lambda = -1 \quad \lambda = -1$$

for $\lambda = -1$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \qquad -x_1 + x_2 = 0$$

$$S_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

for $\lambda = 1$

$$\begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \qquad S_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$c_1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-1)^n + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot 1^n$$

(6)

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-1)^n + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot 1^n$$

for $n = 0$

$$\begin{bmatrix} 10 \\ 40 \end{bmatrix} =$$

$$10 = c_1 + c_2$$

$$40 = c_1 + 3c_2$$

$$30 = 2c_2$$

$$c_2 = 15$$

$$c_1 = -5$$

for $n = 1$

$$-5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot -1 + 15 \begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot 1$$

$$-5 + 15 = 10$$

$$-5 + 45 = 40$$

$$\text{tank}_1 = 10$$

$$\text{tank}_2 = 40$$

$$\frac{10}{50} = \%20$$

(7)